

Shadow Banking and the Four Pillars of Traditional Financial Intermediation

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Roadmap

1. Shadow banking: what & why
2. Aim & methodology
3. The model
4. Solving the model & welfare
5. How complementarity & the quadrilogy emerge
6. Conclusion

What is shadow banking?

Traditional (retail) bank = four bundled pillars:

- **SME lending** — credit to “special borrowers” needing close oversight
- **Insured deposit taking** — safe, liquid claims for “special depositors”
- **Lender of last resort (LOLR)** — public liquidity in bad times
- **Prudential supervision** — the price of admission

Shadow banking = credit intermediation *outside* this regulated bundle: money-market funds, investment banks, conduits, wealth-management vehicles.

The research question

*Why are SME lending, insured deposits, LOLR, and supervision bundled together?
Should that “quadrilogy” be reconsidered now that shadow banks invade core territory?*

Position in the literature: the paper is closest to regulatory arbitrage, but gives a normative mechanism-design account of the regulated/shadow split without assuming comparative advantage.

Strategy: derive the bundle as the equilibrium of a game between banks and a government.

- Government capability: a *monopoly on taxation* $\Rightarrow$ unique ability to create liquidity in tail events.
- Public insurance (LOLR, DI) is *costly* (deadweight cost of taxation / political opprobrium).
- Supervision lowers that cost by curbing moral hazard $\Rightarrow$ **complementarity**.

Methodology: optimal mechanism design

The government designs an optimal **mechanism**, not a single policy instrument.

- ① Government offers a regulatory **contract** $\{\tau_0, k, l, m, \{d_{\omega_1}\}\}$.
- ② Banks **participate** or take the outside option (pure shadow banking).
- ③ Government **maximizes welfare** W subject to the banks' participation constraint, which pins down the transfer τ_0 .

Four assumptions to flag orally

Special depositors want safe assets; private liquidity is scarce; public funds are costly; bankers choose leverage under limited liability.

Environment: two dates, three players

Two periods $t = 0, 1$, one good. Players: **investors**, **bankers** $i \in [0, 1]$, **government**.

At $t = 1$ an aggregate state $\omega = (\omega_1, \omega_2)$ realizes:

- **Fiscal state** $\omega_1 \in \{G, B\}$ — sets the shadow cost of public funds $1 + \lambda_{\omega_1}$, with

$$0 \leq \lambda_G < \lambda_B. \quad (\text{verifiable})$$

- **Liquidity state** $\omega_2 \in \{g, b\}$ — sets bank revenue r_{ω_2} , with

$$r_g = 2 > r_b = 0. \quad (\text{not verifiable})$$

Probabilities: $p_\omega \equiv p_{\omega_1} p_{\omega_2|\omega_1}$ (e.g. p_{Gg}).

Two frictions: no store of value; investors *cannot commit* future funds (Holmström–Tirole 1998). $\Rightarrow$ Only the *government* can supply liquidity in bad times.

Players I — Investors

Ordinary investors: risk-neutral, no discounting, deep pockets.

$$\mathbb{E}_\omega[c_0 + c_{1,\omega}]$$

Uninsured in equilibrium.

Special depositors (mass $\mu \leq 1$): *infinitely risk-averse at $t = 0$* (Epstein–Zin).

- Willing to pay $1 + \theta$ (with $\theta > 0$) at $t = 0$ for a *certain* unit at $t = 1$.
- Demand **absolutely safe** claims — impossible without government backing.

Implication

Because banks face an *aggregate* shock, they cannot manufacture safety alone $\Rightarrow$ a role for **deposit insurance**. θ is the social surplus from safe-asset creation.

Players II — Bankers: the choice variable

Mass 1 of bankers, protected by **limited liability**. Each must finance a unit “project” (the SME) at $t = 1$; the banker is *indispensable*.

Date-0 choice = leverage. Bank issues uninsured debt with face value $\{d_{\omega_1}\}$, contingent on ω_1 only. Repayment capped by limited liability:

$$\text{actual repayment} = \min\{d_{\omega_1}, r_{\omega_2}\}.$$

May also issue μ^i **insured deposits** — but only *with the government's approval*.

Banker's payoff: values $t = 0$ consumption and the non-pledgeable private benefit $b > 1$ of investing:

$$U \equiv \mathbb{E}_{\omega} [c_0^i + b x_{\omega}^i], \quad x_{\omega}^i \in \{0, 1\}.$$

Leverage mechanism: debt issuance raises date-0 cash but increases the probability of illiquidity at $t = 1$.

Players III — The government

Two instruments at $t = 0$:

- **Monitor** (regulate) bank i : set its leverage $\{d_{\omega_1}\}$ (capital requirement), at fixed cost $c \geq 0$.
- **Tax & insure**: raise revenue at $t = 1$ to fund bailouts / honor LOLR & DI. Collecting 1 costs $1 + \lambda_{\omega_1}$.

Preferences & the key parameter β . Weight 1 on investors; total weight β on a *project being implemented* (the SME stakeholders). The crucial assumption:

$$1 + \lambda_B > \beta > 1 + \lambda_G$$

- In state G: $\beta > 1 + \lambda_G \Rightarrow$ government *cannot resist* bailing out (discretionary).
- In state B: $\beta < 1 + \lambda_B \Rightarrow$ government *would like to renege* — only a $t = 0$ commitment (LOLR) sustains rescue.

Bailout vs. insurance — the central distinction

Bailout

Ex-post, time-inconsistent rescue. Government *wishes* it could commit not to, but can't resist once the bank is illiquid (state G).

Public insurance (LOLR / DI)

Ex-ante contractual commitment the government *wants*: it monetizes its unique liquidity-provision ability and pulls the bank into the regulated net.

Notation used later: expected deadweight cost per unit of public transfer,

$$\Lambda_{\omega_2} \equiv \sum_{\omega_1 \in \{G, B\}} p_{\omega_1} p_{\omega_2 | \omega_1} \lambda_{\omega_1}, \quad \bar{\lambda} \equiv \Lambda_g + \Lambda_b = p_G \lambda_G + p_B \lambda_B.$$

Λ_g is the expected fiscal cost when revenue is high; Λ_b when revenue is low.

Timing



Date 0

- ① Government offers contract $\{\tau_0, k, l, m, d_{\omega_1}\}$
- ② Banks accept *or* become shadow banks
- ③ Each bank chooses leverage $\{d_{\omega_1}\}$

Date 1

- ① State ω realized; revenue r_{ω_2}
- ② Debt repayment $\min\{d_{\omega_1}, r_{\omega_2}\}$
- ③ Investment $x_{\omega}^i \in \{0, 1\}$
- ④ Depositor repayment
- ⑤ Government inspects & rescues: bailout (state G) or honors LOLR/DI (state B)

Timing implication: debt repayment and investment precede depositor repayment and public rescue, so liquidity shortfalls are realized before insurance is paid.

Welfare components

Policy evaluation compares total surplus W across feasible policy packages.

- **Investment surplus:** continuation creates private benefit b and social value β , but uses one unit of resources.
- **Revenue:** high-liquidity banks have $r_g = 2$; low-liquidity banks have $r_b = 0$.
- **Safe-asset surplus:** insured deposits create value θ for special depositors.
- **Public-fund cost:** bailouts, LOLR and DI require taxation with deadweight cost λ_{ω_1} .
- **Monitoring cost:** regulation costs c , but can reduce future public-fund costs.

Transferable utility implication

The date-0 transfer τ_0 only moves surplus between the government and banks, so solve for W first and use τ_0 only to satisfy participation.

Full welfare object

Bank i 's utility:

$$U_i \equiv \mathbb{E}_\omega \left[bx_\omega^i + \min\{d_{\omega_1}^i, r_{\omega_2}\} + (1 + \theta)\mu^i + \tau_0^i \right].$$

Total government + bank welfare:

$$W \equiv \mathbb{E}_\omega \int_0^1 \left[(b + \beta - 1)x_\omega^i + r_{\omega_2} + \theta\mu^i - cm^i - \lambda_{\omega_1} \max\{0, x_\omega^i + \mu^i + \min\{d_{\omega_1}^i, r_{\omega_2}\} - r_{\omega_2}\} \right] di.$$

Date-0 transfers τ_0^i do not enter W because utility is transferable at date 0.

The policy object & the outside option

Government chooses a single one-size-fits-all contract

$$\{\tau_0, k, l, m, \{d_{\omega_1}\}\},$$

- $k = 1$: deposit insurance $l = 1$: LOLR $m = 1$: regulation d_{ω_1} : prescribed leverage (enforced iff $m = 1$).

Investment indicator $x_{\omega}^i = 1$ iff there is enough cash — either from revenue, or a rescue:

$$x_{\omega}^i = 1 \iff r_{\omega_2} \geq d_{\omega_1} + \mu^i + 1, \text{ or } (l = 1 \text{ (LOLR) or } \omega_1 = G \text{ (bailout)}).$$

Outside option

Pure shadow banking is *always available* ($k = l = m = \tau_0 = 0$). The optimal mechanism solves

$$\max_{\{k, l, m, d_{\omega_1}\}} W(k, l, m, \{d_{\omega_1}\}) \quad \text{s.t. participation (which fixes } \tau_0).$$

Solution strategy

Exploit transferable utility:

- Bank's utility is driven to its reservation value U^{SB} (shadow-bank outside option).
- Government's payoff = $W(k, l, m, \{d_{\omega_1}\}) - U^{\text{SB}}$.
- τ_0 chosen to make the bank *indifferent* to going shadow.

It is sufficient to compute W under each configuration (k, l, m) and select the welfare-maximizing arrangement.

Plan

(1) Shadow banking $m = 0 \rightarrow$ (2) Regulated banking $m = 1 \rightarrow$ (3) Compare $\Rightarrow$ complementarity & the quadrilogy.

Pure shadow banking: $m = 0, k = 0, l = 0$

Leverage. In fiscal state G, the bank anticipates a bailout if illiquid, so $d_G = 2$. In fiscal state B, no LOLR is available; comparing $d_B = 1$ gives $1 + b$ in state (B, g) while $d_B = 2$ gives 2, so $b > 1$ implies $d_B = 1$.

$$s \equiv b + \beta - 1, \quad x_{Gg} = x_{Gb} = x_{Bg} = 1, \quad x_{Bb} = 0.$$

$$\begin{aligned} U^{SB} &= p_{Gg}(2 + b) + p_{Gb}b + p_{Bg}(1 + b) \\ &= (1 - p_{Bb})b + 2p_{Gg} + p_{Bg}, \end{aligned}$$

$$\begin{aligned} W^{SB} &= p_{Gg}(2 + s - \lambda_G) + p_{Gb}(s - \lambda_G) + p_{Bg}(2 + s) \\ &= (1 - p_{Bb})s + 2p_g - \lambda_G p_G. \end{aligned}$$

- The factor $1 - p_{Bb}$ is the continuation probability: only (B, b) has no investment.
- The bailout deadweight loss is paid in fiscal state G, with expected cost $\lambda_G p_G$.

Shadow banking + LOLR: $m = 0, k = 0, l = 1$

With LOLR, the shadow bank chooses $d_G = d_B = 2$; public liquidity in B induces empty-shell behavior.

State changes relative to W^{SB}

- (B, g): investment occurs both with and without LOLR, but LOLR now pays one unit because $d_B = 2$; welfare change = $-\lambda_B$.
- (B, b): no investment without LOLR; LOLR permits investment with transfer one; welfare change = $s - \lambda_B$.

Expected welfare gain

$$\begin{aligned} W^{SBL} - W^{SB} &= p_{Bg}(-\lambda_B) + p_{Bb}(s - \lambda_B) \\ &= p_{Bb}s - p_B\lambda_B \\ &= p_{Bb}(b + \beta - 1) - p_B\lambda_B. \end{aligned}$$

Condition

LOLR is desirable in shadow banking iff $p_{Bb}(b + \beta - 1) > p_B\lambda_B$.

Shadow banking + DI: $m = 0, k = 1, l = 0$

Special deposits create surplus θ per unit, but an unregulated shadow bank can become an empty shell when deposits are due. Deposit insurance therefore pays the full insured amount in all fiscal states.

$$\begin{aligned}\bar{\lambda} &\equiv p_G \lambda_G + p_B \lambda_B = \Lambda_g + \Lambda_b, \\ W^{\text{SBD}} - W^{\text{SB}} &= \underbrace{\mu \theta}_{\text{safe-deposit surplus}} - \underbrace{\mu \bar{\lambda}}_{\text{full expected fiscal cost of DI}} \\ &= \mu(\theta - \bar{\lambda}).\end{aligned}$$

- The key difference from regulated DI is that revenue in state g does not protect taxpayers: the bank can choose leverage so deposits are publicly repaid.
- DI is desirable in shadow banking iff $\theta > \bar{\lambda}$.

Pure regulated banking: $m = 1, k = 0, l = 0$

Regulation constrains leverage to $d_G = d_B = 1$ and costs c .

Only state (G, g) changes

- Pure shadow banking: $d_G = 2$, so the bank is an empty shell after debt repayment and receives bailout one.
- Regulated banking: $d_G = 1$, so $r_g - d_G = 1$ and investment is privately financed.
- Avoided deadweight loss in (G, g) : λ_G .

Expected welfare gain

$$W^{\text{RB}} - W^{\text{SB}} = \underbrace{p_{Gg}\lambda_G}_{\text{avoided bailout cost}} - \underbrace{c}_{\text{monitoring cost}},$$

$$W^{\text{RB}} = W^{\text{SB}} + p_{Gg}\lambda_G - c.$$

Interpretation

Regulation has direct value even without LOLR or DI because it prevents inefficient discretionary bailout in (G, g) .

Regulated banking + LOLR: $m = 1, k = 0, l = 1$

Regulation prevents the LOLR-induced choice $d_B = 2$; the bank remains at $d_B = 1$.

State changes relative to W^{RB}

- (B, g): the bank self-finances because $r_g - d_B = 1$; LOLR does not pay.
- (B, b): no investment without LOLR; with LOLR, investment occurs and public transfer is one.
- Welfare change in (B, b): $s - \lambda_B$.

Expected welfare gain

$$\begin{aligned} W^{RBL} - W^{RB} &= p_{Bb}(s - \lambda_B) \\ &= p_{Bb}[b + \beta - (1 + \lambda_B)]. \end{aligned}$$

Complementarity

$$(W^{RBL} - W^{RB}) - (W^{SBL} - W^{SB}) = p_{Bg}\lambda_B > 0.$$

Regulated banking + DI: $m = 1, k = 1, l = 0$

Regulation sets leverage to $d_G = d_B = 1 - \mu$, so in good-liquidity states the bank can both invest and repay special depositors:

$$d + \mu + 1 = (1 - \mu) + \mu + 1 = 2 = r_g.$$

State-by-state DI effect

- g states: safe-deposit surplus $\mu\theta$, no public DI payment.
- (G, b): DI pays μ ; welfare change $\mu\theta - \mu\lambda_G$.
- (B, b): DI pays μ ; welfare change $\mu\theta - \mu\lambda_B$.

Expected welfare gain

$$\begin{aligned} W^{\text{RBD}} - W^{\text{RB}} &= \mu\theta - \mu(p_{Gb}\lambda_G + p_{Bb}\lambda_B) \\ &= \mu(\theta - \Lambda_b). \end{aligned}$$

Complementarity

$$(W^{\text{RBD}} - W^{\text{RB}}) - (W^{\text{SBD}} - W^{\text{SB}}) = \mu\Lambda_g > 0.$$

Key net-benefit formulas

Let $s \equiv b + \beta - 1$. The six cases imply:

$$W^{\text{SBL}} - W^{\text{SB}} = p_{\text{Bb}}s - p_{\text{B}}\lambda_{\text{B}},$$

$$W^{\text{RB}} - W^{\text{SB}} = p_{\text{Gg}}\lambda_{\text{G}} - c,$$

$$W^{\text{RBD}} - W^{\text{RB}} = \mu(\theta - \Lambda_{\text{b}}).$$

$$W^{\text{SBD}} - W^{\text{SB}} = \mu(\theta - \bar{\lambda}),$$

$$W^{\text{RBL}} - W^{\text{RB}} = p_{\text{Bb}}(s - \lambda_{\text{B}}),$$

Positive interaction terms

$$(W^{\text{RBL}} - W^{\text{RB}}) - (W^{\text{SBL}} - W^{\text{SB}}) = p_{\text{Bg}}\lambda_{\text{B}} > 0, \quad (W^{\text{RBD}} - W^{\text{RB}}) - (W^{\text{SBD}} - W^{\text{SB}}) = \mu\Lambda_{\text{g}} > 0.$$

Welfare decomposition

Using the net-benefit terms above,

$$W(k, l, m) = W^{\text{SB}} + [p_{\text{Bb}}s - p_{\text{B}}\lambda_{\text{B}}]l + \mu(\theta - \bar{\lambda})k + (p_{\text{Gg}}\lambda_{\text{G}} - c)m \\ + p_{\text{Bg}}\lambda_{\text{B}}ml + \mu\Lambda_{\text{g}}mk.$$

Equivalently,

$$W(k, l, m) = W^{\text{SB}} + (W^{\text{SBL}} - W^{\text{SB}})l + (W^{\text{SBD}} - W^{\text{SB}})k + (W^{\text{RB}} - W^{\text{SB}})m \\ + [(W^{\text{RBL}} - W^{\text{RB}}) - (W^{\text{SBL}} - W^{\text{SB}})]ml + [(W^{\text{RBD}} - W^{\text{RB}}) - (W^{\text{SBD}} - W^{\text{SB}})]mk.$$

Core implication

The positive ml and mk terms are the algebraic source of complementarity between regulation and public insurance.

Quadrilogy conditions: insurance only with regulation

A traditional banking system requires LOLR and DI to be valuable under regulation, but not as standalone supports for shadow banks.

Deposit insurance

$$W^{\text{RBD}} - W^{\text{RB}} > 0 \iff \theta > \Lambda_b, \quad W^{\text{SBD}} - W^{\text{SB}} < 0 \iff \theta < \bar{\lambda}.$$

Hence:

$$\Lambda_b < \theta < \bar{\lambda}.$$

LOLR

$$W^{\text{RBL}} - W^{\text{RB}} > 0 \iff s > \lambda_B, \quad W^{\text{SBL}} - W^{\text{SB}} < 0 \iff p_{Bb}s < p_B \lambda_B.$$

Equivalently:

$$0 < p_{Bb}[b + \beta - (1 + \lambda_B)] < p_{Bg} \lambda_B.$$

Quadrilogy conditions: regulation cost threshold

With regulation, LOLR and DI, welfare relative to pure shadow banking is

$$\begin{aligned} W^{\text{RBLD}} - W^{\text{SB}} &= (W^{\text{RB}} - W^{\text{SB}}) + (W^{\text{RBL}} - W^{\text{RB}}) + (W^{\text{RBD}} - W^{\text{RB}}) \\ &= p_{\text{Gg}}\lambda_{\text{G}} - c + p_{\text{Bb}}[b + \beta - (1 + \lambda_{\text{B}})] + \mu(\theta - \Lambda_{\text{b}}). \end{aligned}$$

Therefore traditional banking is preferred to pure shadow banking iff

$$c \leq c^* = p_{\text{Gg}}\lambda_{\text{G}} + p_{\text{Bb}}[b + \beta - (1 + \lambda_{\text{B}})] + \mu(\theta - \Lambda_{\text{b}}).$$

Conclusion

Under the LOLR, DI and regulation-cost inequalities, SME lending, insured deposits, LOLR and supervision are jointly optimal rather than separable policy instruments.

Conclusion

- **Method:** an optimal mechanism-design game between banks and government, solved via transferable utility — compute W , back out τ_0 from participation.
- **Core result:** the welfare decomposition (7) has *positive interaction terms* $\Rightarrow$ regulation and public insurance (LOLR, DI) are **complements** — economies of scope in supervision.
- **Consequence:** the four-pillar “quadrilogy” is *derived*, not assumed; underpriced insurance defends the regulated perimeter against migration.

No comparative advantage is posited — the structure of banking is an equilibrium outcome.

Thank you

Backup: literature interpretations

Breathing room

Regulation stifles lending; shadow banking undoes a *state failure* (Ordoñez 2018).

Regulatory arbitrage

Evades capital requirements yet still gets bailed out — a *nuisance* (Acharya et al.; Plantin 2015).

Neglected risk / comparative adv.

False “safe” assets (Gennaioli et al.) or genuine specialization (Hanson et al. 2015).

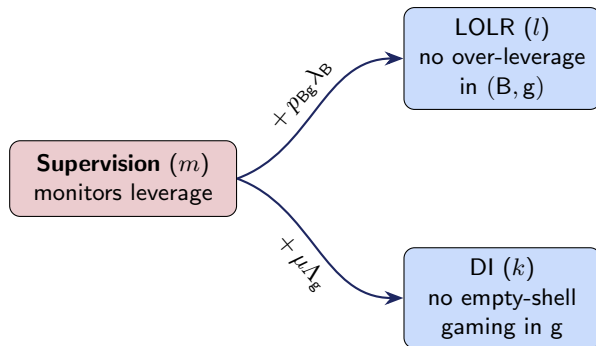
Where this paper sits

Closest to regulatory arbitrage, but normative: it derives the sectors as an equilibrium outcome.

Backup: load-bearing assumptions

- **Special depositors** value absolutely safe claims: drives deposit insurance.
- **No store of value + scarce private commitment** makes government liquidity unique.
- $1 + \lambda_B > \beta > 1 + \lambda_G$ gives bailouts in G, but makes LOLR commitment valuable in B.
- **Moral hazard is leverage choice** under limited liability: supervision has a clear target.
- **Shadow banking remains available** as an outside option, so migration constrains the mechanism.

Backup: economies of scope in supervision



The *same* monitoring infrastructure that curbs leverage caps the cost of *both* taxpayer put options. One fixed cost c , two benefits $\Rightarrow$ increasing returns to scope.

Backup: coexistence of the two sectors

Add **heterogeneous monitoring costs**: a fraction α of banks have finite cost c , while the rest are prohibitively hard to supervise.

- Fraction α : regulated, with LOLR and DI.
- Fraction $1 - \alpha$: shadow, with no public support.
- Special depositors are served by regulated banks because DI has lower fiscal cost under supervision.

Underpriced public insurance supports retention of supervisable banks inside the regulated perimeter.